# Ganesh Gupta

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## Experience

**Oracle** | *Software Engineer, AI Platform* Jun 2025 – Present

- Led a production **multi-agent system** on **OpenAI Agents SDK**, **LangGraph**, and **MCP** tool use that autonomously processes **financial workflows**, handling **millions of daily transactions** at **99%+ uptime**.
- Built agent workflows across **GPT, Claude, Llama, and Gemini** with **CrewAI** and **AutoGen** alongside LangGraph, covering **tool calling**, memory, and multi-agent planning.
- Built **evaluation harnesses** measuring LLM output quality, safety, **hallucination rates**, and grounding across agent versions, gating releases with **regression suites**.
- Implemented **GraphRAG** on a **Neo4j knowledge graph** in production, improving grounding over vector RAG on complex **multi-hop** queries.

**QureAI** | *AI Engineer* Mar 2026 – Present

- Raised **top-3 retrieval recall from 61% to 89%** on a 2K-query eval set by rebuilding the pipeline with **hybrid dense-sparse retrieval** and **re-ranking**.
- Built an SRE incident agent on the **Claude Agent SDK** that hands data questions to a **SQL agent** over **A2A**, with an eval loop grading its root-cause analyses against a **golden set**.
- Built a **DAG-generation agent** with **30+ MCP tools** that turns a requirements questionnaire and a base DAG into a validated, published workflow, for a **regulated healthcare** product.

**Nomura Research Institute Financial Technologies** | *Software Engineer* Sep 2019 – Aug 2023

- Architected and operated **25+ production Python and Java microservices** for **1M+ users** in **financial services** at **99.9% uptime**.
- Cut nightly batch aggregation from **8 minutes to 90 seconds** by optimizing **SQL and PL/SQL** on a financial RDBMS with indexed views and partitioning.
- Cut model regressions **28%** and validation time from **6 hours to 90 minutes** with an automated **evaluation and release-gating** framework.
- Led **architecture and code reviews** and mentored engineers on system design, reliability, and secure API practices.

## Projects

**RootCause AI** `github.com/ganeshhgupta/rootcause-ai`

- **LangGraph** multi-agent system that routes production alerts to specialized sub-agents via **tool-use** chains and writes structured root-cause reports, cutting **MTTR 65%**.

**LangFetch** `tinyurl.com/LangFetch`

- **LangGraph** agent that decomposes natural-language **financial queries** into dependency-ordered **SQL** subqueries with **Pinecone** retrieval, served via **FastAPI** as an **MCP** server.

**Watch-your-LLM** `github.com/ganeshhgupta/watch-your-llm`

- **LLM evaluation** platform with rubric-based scoring, **hallucination detection**, regression alerting, a **FastAPI** backend, and a **React/TypeScript** dashboard.

## Education

**University of Texas at Arlington** | *M.S. Computer Science, GPA 4.0/4.0* Aug 2023 – May 2025

**MCKV Institute of Engineering** | *B.Tech Computer Science* Aug 2016 – Jul 2020

## Skills

**Languages:** Python, Java, SQL, TypeScript

**Agent Memory and Retrieval:** Knowledge Graphs (Neo4j), GraphRAG, Hybrid Dense-Sparse Retrieval, Re-Ranking, Context Management, Agent Memory, Embeddings

**ML and Evaluation:** PyTorch, Fine-Tuning, LLM Evaluation, Hallucination Detection, Regression Suites, Claude Agent SDK, LangGraph, MCP, AWS Bedrock

**Certifications:** AWS Certified Developer Associate, GCP Professional MLOps Engineer